

LESSON 110 (0.5 GROUND and 1.0 AATD)

Lesson Objective

During this lesson, the applicant will become familiar with the fundamentals of flight in the training device. This will include demonstrating various turns, climbs and descents with various power settings. This will enhance the applicants' understanding of how to enter and level off from climbs and descents as well as how to roll out from turns. The applicant will also review previously learned maneuvers for proficiency.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the readings as needed.
 - Chapter 3 of the Airplane Flying Handbook - Basic Flight Maneuvers.
 - Level turns.
 - Climbs and climbing turns.
 - Descents and descending turns.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 03 - Basic Flight Maneuvers.

Instructor Guidance

- This is the third simulator lesson and the first time the student is introduced to climbs, descents, and turns. Keep the ground portion concise and focused — the student will retain these procedures far better by doing them in the sim than by listening to lengthy explanations.
- During the ground brief, quiz the student on their AFH Chapter 3 readings before introducing new content. If they haven't read it, address that expectation clearly — prepared students get more out of simulator time.
- When introducing each maneuver in the sim, follow a consistent pattern: brief explanation, demonstrate once, then hand controls to the student. Avoid over-demonstrating. Students at this stage learn by doing, and instructor demonstrations that go on too long reduce the student's stick time without meaningful benefit.
- Trim is the most common deficiency at this stage. If the student is muscling the aircraft through climbs and descents rather than trimming, stop and correct it immediately. Reinforce "trim to relieve pressure" as a non-negotiable habit every single time a power or pitch change is made.
- When teaching the level-off from climbs and descents, emphasize the 10% VSI anticipation rule early. Students who wait until they reach the target altitude before starting the level-off

will consistently overshoot. Repetition in the sim builds the timing instinct needed for the aircraft.

- For medium level turns, watch closely for students who increase bank without adding back pressure. Let them experience the altitude loss first, then explain why it happened — this is more effective than preventing the error before it occurs.
- Emphasize the traffic scan at every level-off from a climb. The requirement to level off every 500 feet to scan for traffic should be stated and reinforced throughout the session, not just mentioned once.
- Do not rush through all the maneuvers if the student is struggling. It is better to solidify climbs and descents before moving to turns than to introduce everything at a surface level. If time runs short, prioritize depth over breadth.
- During the post-brief, have the student self-critique their trim usage specifically — this forces self-awareness on the habit that will be most critical going forward.

Lesson Content (Ground)

Climbing

- Review how to enter a climb, maintain a climb, and level off from a climb.
- Climbs:
 - Entry:
 - Power.
 - Pitch.
 - Rudder.
 - Trim.
 - Maintaining:
 - Lookout.
 - Attitude indicator.
 - Airspeed indicator.
 - Altimeter.
 - HSI.
 - Maintain coordination as needed.
 - Leveling off:
 - Anticipate by 10% of VSI.
 - Pitch.
 - Power.
 - Trim.

Descending

- Descents:
 - Entry:
 - Power.
 - Pitch.
 - Rudder.
 - Trim.
 - Maintaining:
 - Lookout.
 - Attitude indicator.
 - Airspeed indicator.
 - Altimeter.
 - HSI.
 - Maintain coordination as needed.
 - Leveling off:
 - Anticipate by 10% of VSI.
 - Pitch.
 - Power.
 - Trim.

Turning

- Medium Level Turns:
 - Entry:
 - Lookout.
 - Bank - ailerons.
 - Coordination - rudder.
 - Back pressure - stabilator.
 - Maintaining:
 - Lookout.
 - Attitude indicator.
 - Airspeed indicator.
 - Altimeter.
 - Coordination.
 - Leveling off:
 - Anticipate by ½ bank angle.
 - Roll the wings level.

Lesson Content (AATD)

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all of the checklists.

Review Previous Exercises as Required

- Use the appropriate lesson plan depending on what needs to be reviewed.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Entering the Climb

- Full power.
- Right rudder.
- Set pitch attitude for the assigned airspeed (use visual references and attitude indicator).
 - V_X climb: about 12.5 degrees.
 - V_Y Climb: about 10 degrees.
 - Cruise climb: about 7.5 degrees.
 - Subject to change based on current conditions.

Maintaining the Climb and Making Corrections

- Keep the desired pitch attitude using TRIM!
- Maintain rudder input as needed.
- Maintain heading using HSI and visual reference.
- Emphasize to level off every 500 feet to scan for traffic.

Leveling Off from a Climb

- Anticipate by using 10 percent of the vertical speed indicator.
- Lower the pitch to straight and level pitch attitude.
- Reduce the power to straight and level power setting (2300 RPM).
- TRIM!

Entering a Descent

- Set power to idle.
- Left rudder.

- Set pitch attitude for a glide descent.
 - Do not let the student chase the airspeed.

Maintaining Glide Descent and Making Corrections

- Keep the desired pitch attitude using TRIM!
- Maintain rudder input as needed.
- Maintain heading using HSI and visual reference.

Leveling Off from a Descent

- Anticipate by using 10 percent of the vertical speed indicator.
- Increase the pitch to straight and level pitch attitude.
- Increase the power to straight and level power setting (2300 RPM).
- TRIM!

Entering Medium Level Turns

- Check for traffic!
- Bank using the ailerons and hold a 30-degree bank (show where that is on the attitude indicator).
- Maintain coordination using the rudder.
- Increase back pressure.
 - More bank → more horizontal component of lift → less vertical component of lift → more back pressure required.

Maintaining Medium Level Turns and Making Corrections

- Maintaining the turn:
 - Lookout.
 - Refer to attitude indicator, airspeed indicator and altimeter.
 - TRIM for the increased back pressure.
- Corrections:
 - Change bank angle as needed.
 - Hold the correction.

Rolling Out from Medium Level Turns

- Anticipate by ½ of the bank angle.
- Roll the wings level - smoothly.
- Decrease pitch attitude.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 111.
- Assign homework:
 - Review climbing, turning, and descending procedures.

Completion Standards

This lesson is complete when the applicant can:

1. Demonstrate proper sequential checklist procedures for engine starting, before taxi, ground check, before takeoff, runway entry, climb, and cruise.
2. Correctly perform previously learned maneuvers with minimal instructor guidance.
3. Demonstrate an understanding of entering and leveling off from climbs and descents as well as how to enter and roll out from turns in the training device.
4. Understand how to control rate of descent when performing a glide descent.
5. Understand the effect of altitude and wind on glide distance.